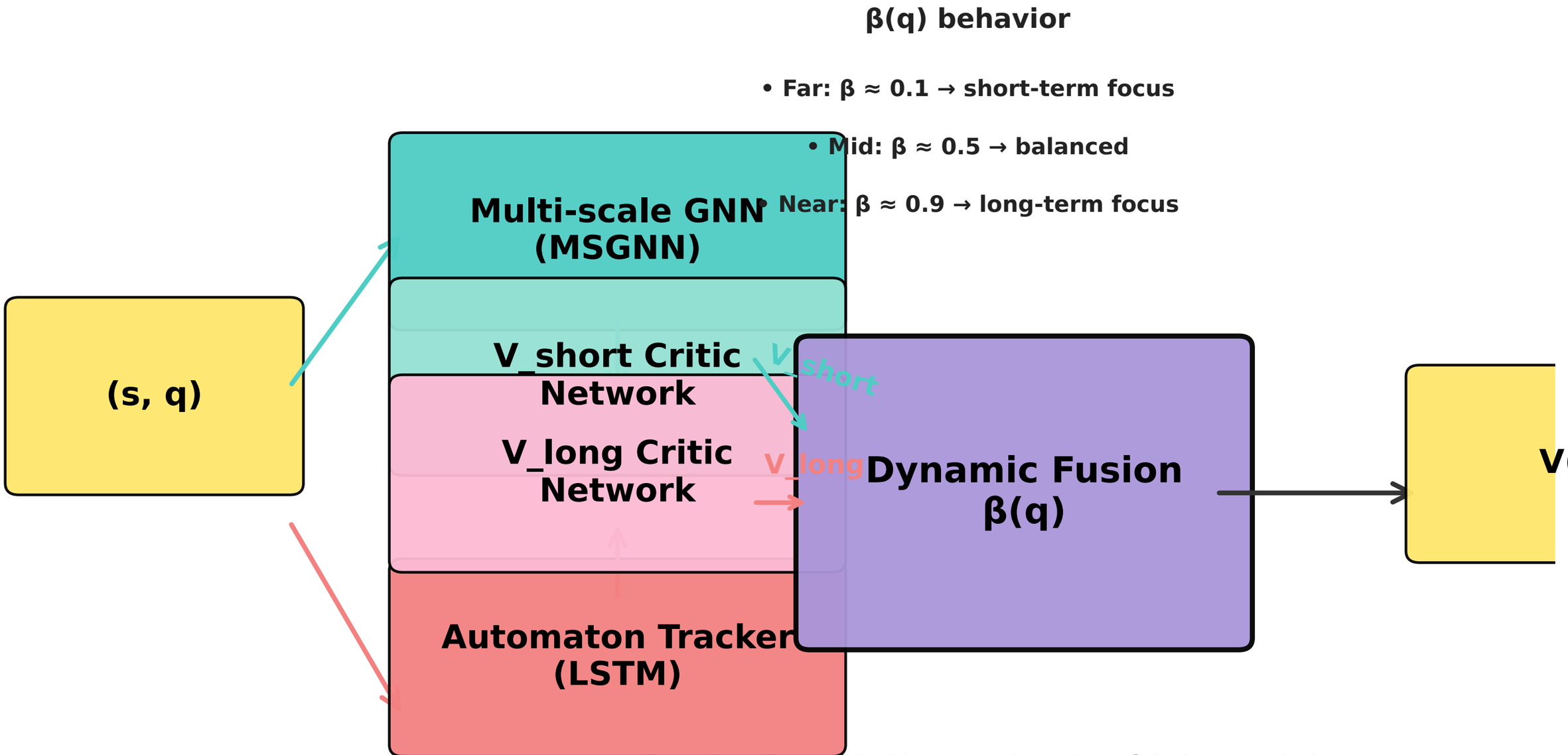
β(q) behavior
• Far: β ≈ 0.1 → short-term focus
• Mid: β ≈ 0.5 → balanced
• Near: β ≈ 0.9 → long-term focus
(s, q)
Multi-scale GNN (MSGNN)
V_short Critic Network
V_long Critic Network
Automaton Tracker (LSTM)
V_short
V_long
Dynamic Fusion β(q)
V(s, q)
V(s, q) = (1 − β(q)) V_short(s, q) + β(q) V_long(q)